

Computational Photography



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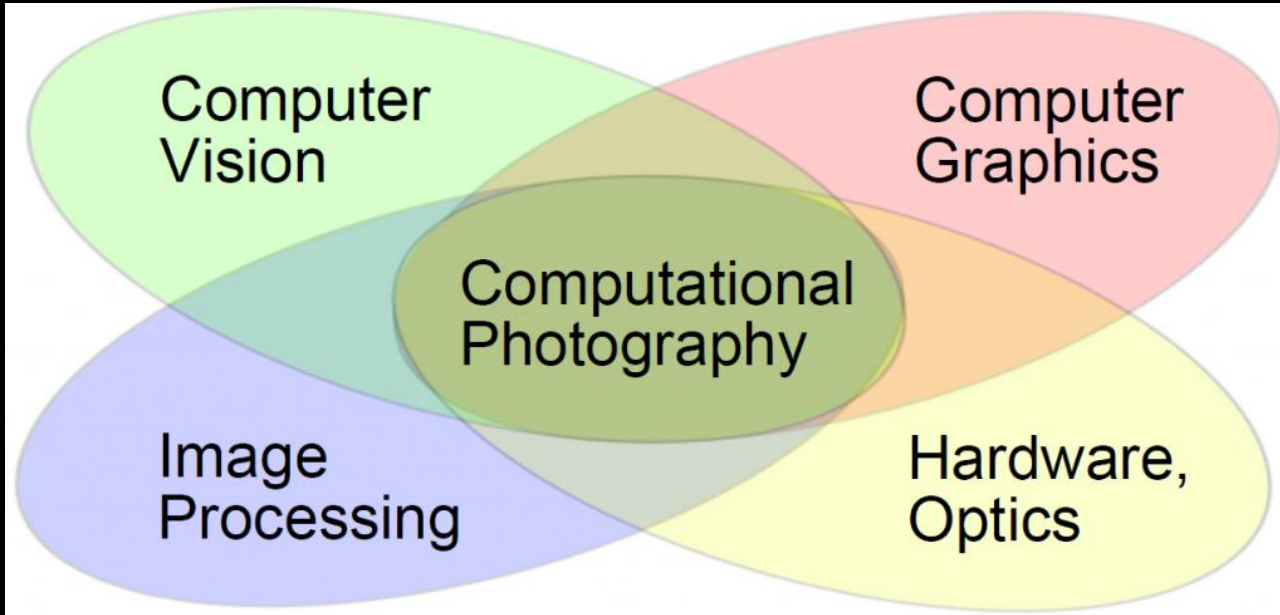
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FEDERICO SANTA MARIA

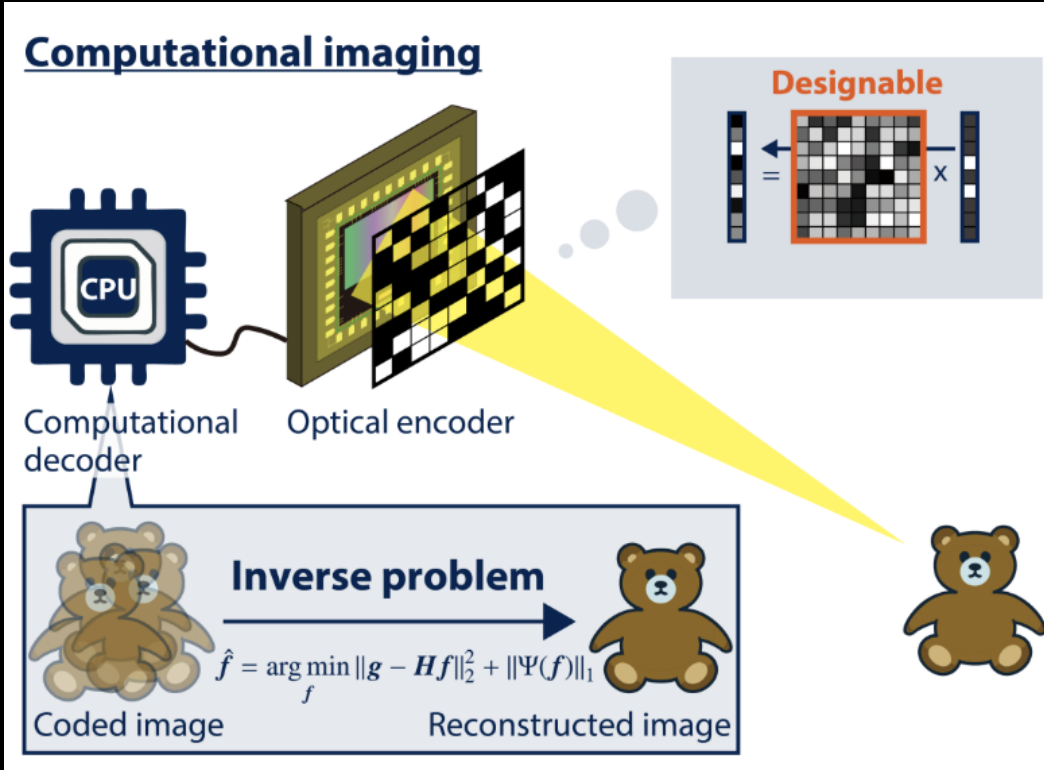
Computational Photography



Areas:

- Computer Science
- Electrical Engineering and Electronics
- Maths
- Physics (optics)

Computational Photography



Goal:

- To capture and recover signal in many dimensions

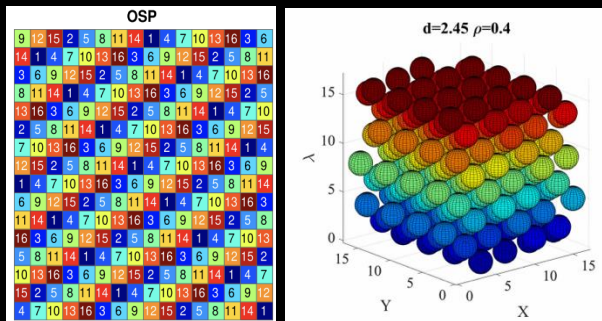
To design

- Hardware
- Software--algorithm to solve the inverse problem--

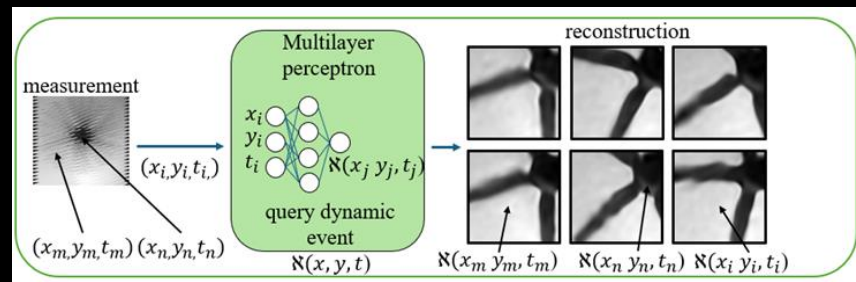
Computational Photography: Applications

- It can be applied to different signals
- It use compressive measurements
- To solve an inverse problem

Multispectral



Video

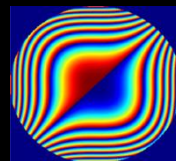


GT phase

Recovered phase

Recovered Amplitude

A woman

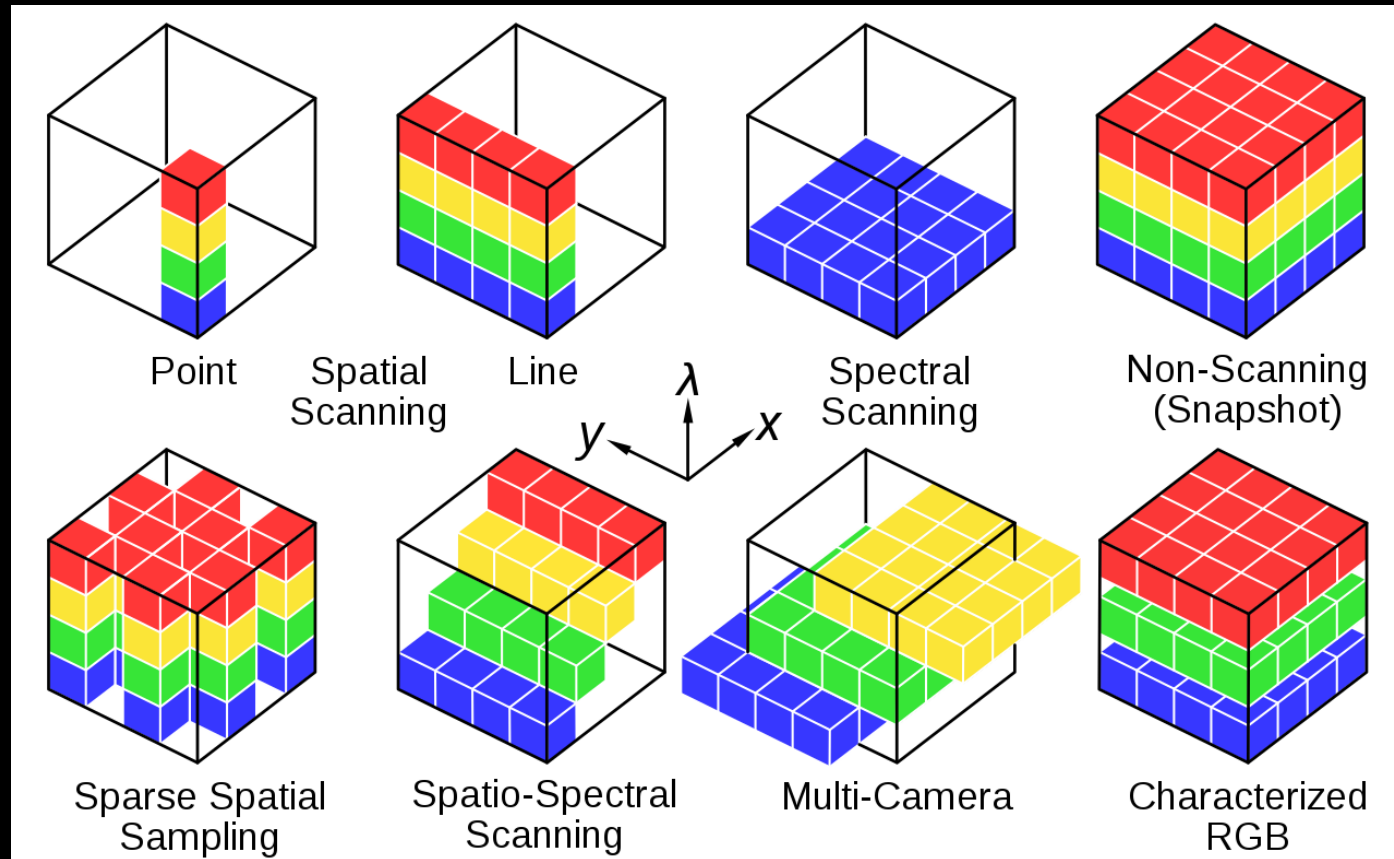


1995
Cubic Phase



Depth

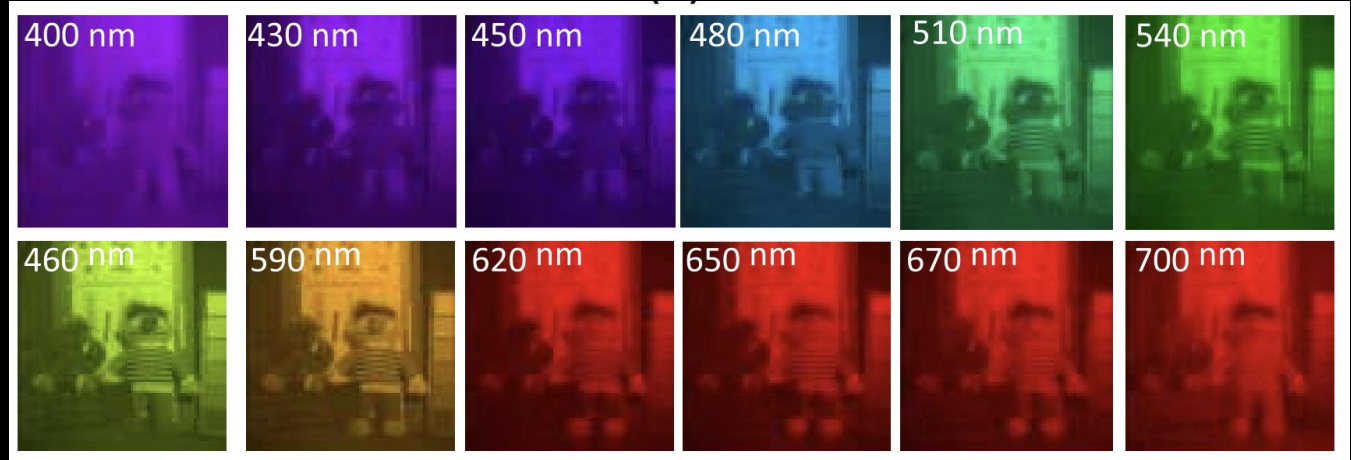
Traditional Approaches to Capture Multispectral Images



Spectral Images

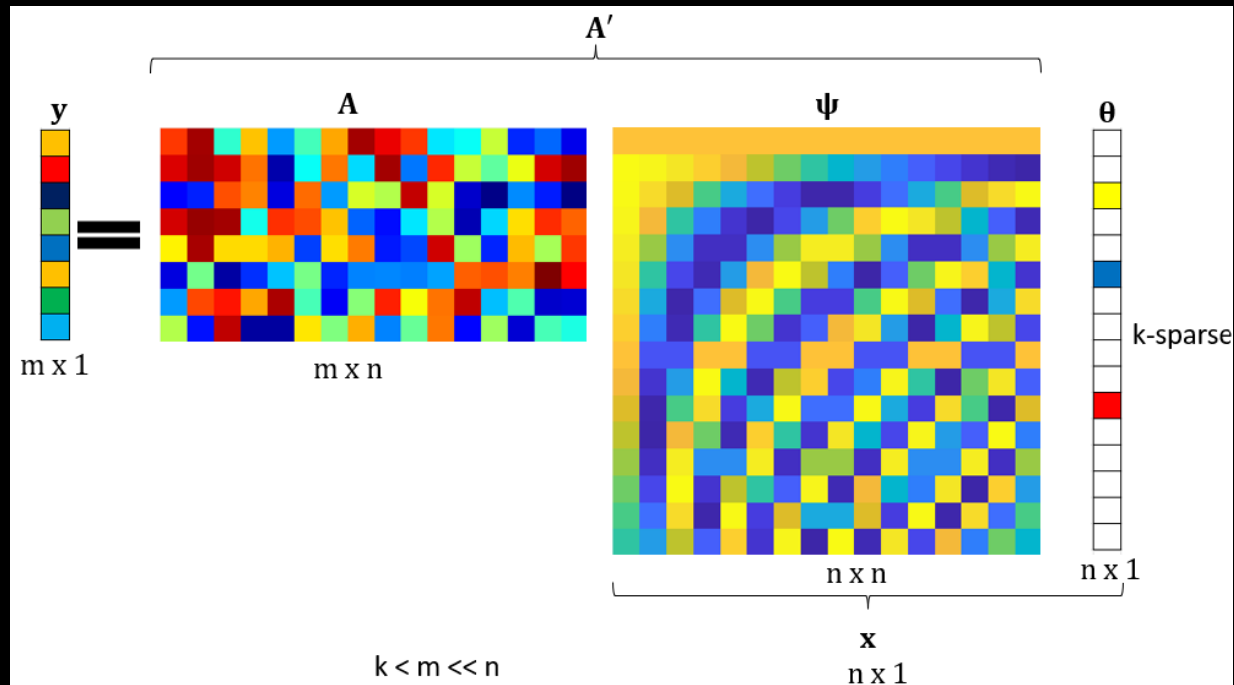


RGB image



Spectral Images in the visible spectrum
from 400 nm to 700 nm

Compressive Sensing

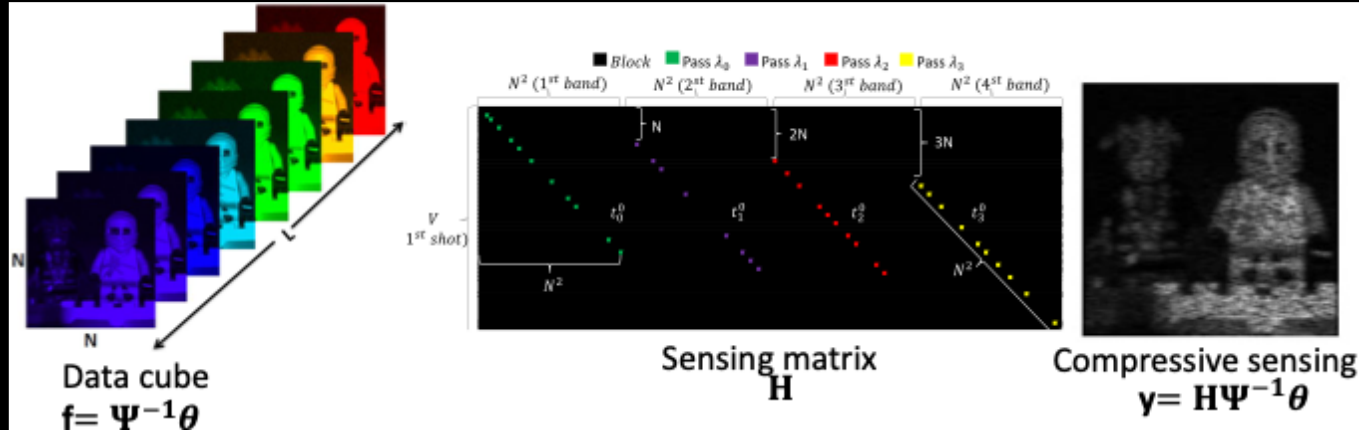


What type of Signal?

- Spectral Images
- Video
- Phase
- MRI, ultrasound
- Text
- Sound
- Any

- The goal is to recover the sparse signal θ from the measurement y
- This is an ill-posed problem

Compressive spectral Imaging



Hardware:
Sensing

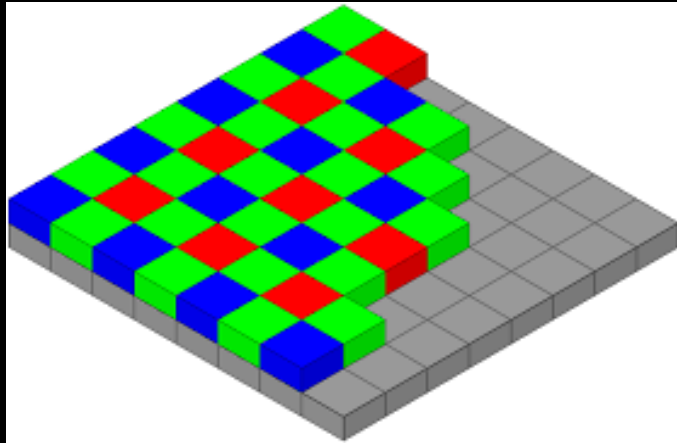
- Underdetermined system of equations



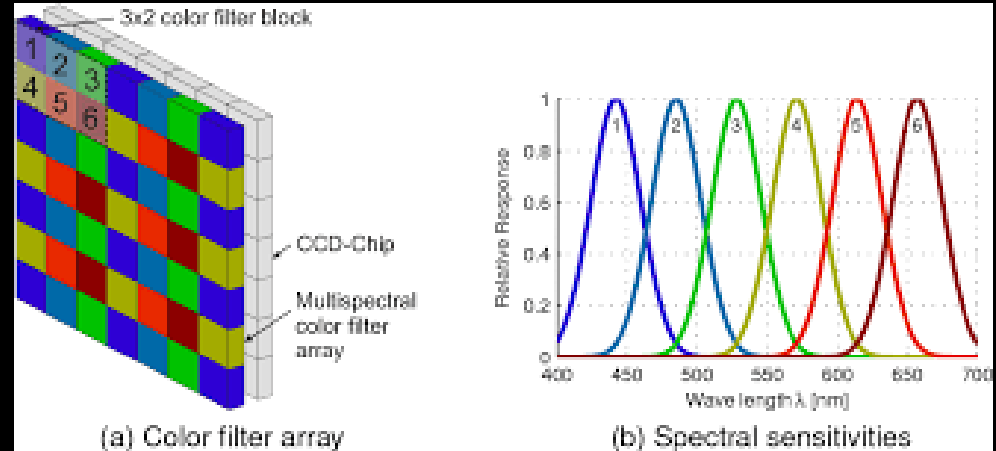
$$\hat{f} = \Psi^{-1}(\operatorname{argmin}_{\theta} \frac{1}{2} \|y - H\Psi\theta\|_2^2 + \tau \|\theta\|_1)$$

Software:
Recovery

Multispectral Filter Array



Bayer filter



Multispectral Filter Array

The recovered signal is multispectral datacube

Random vs Sphere Packing

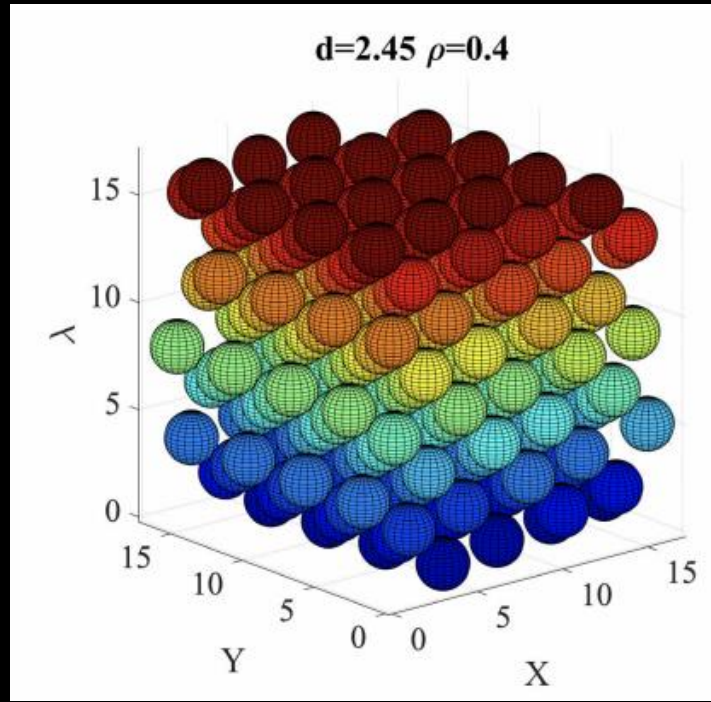
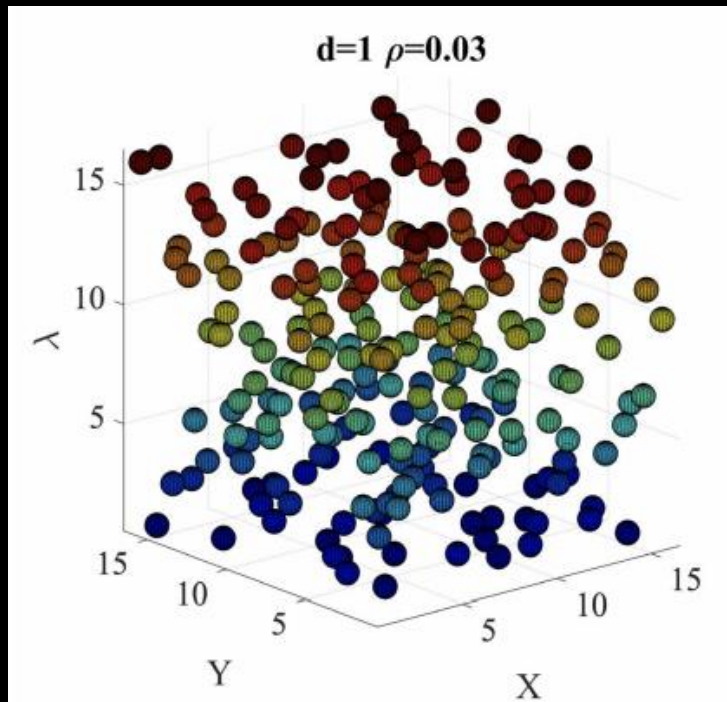
RND

4	5	2	9	8	11	13	7	14	6	10	15	12	3	1	16
5	11	14	13	2	10	9	15	6	8	7	1	4	3	12	16
16	7	1	6	14	8	11	13	3	2	15	4	10	9	5	12
16	9	5	4	10	7	1	6	12	8	3	2	15	11	13	14
9	7	13	15	10	3	12	2	1	16	14	6	8	4	5	11
15	2	4	11	13	1	16	6	14	5	10	9	7	12	8	3
14	7	11	9	6	4	3	2	15	8	13	16	10	12	5	1
7	15	5	11	10	14	13	4	8	9	12	16	3	2	6	1
2	6	1	5	9	11	3	4	16	8	13	10	14	7	12	15
7	15	10	12	1	9	16	11	3	8	6	13	14	5	2	4
6	12	2	14	7	4	1	15	11	5	16	10	13	8	3	9
11	8	12	14	15	7	1	10	13	4	3	9	16	2	5	6
5	2	13	3	15	1	9	6	7	12	8	14	10	11	16	4
6	9	2	15	7	13	16	12	10	4	1	8	11	14	3	5
7	1	12	13	16	3	2	8	14	11	6	15	5	9	4	10
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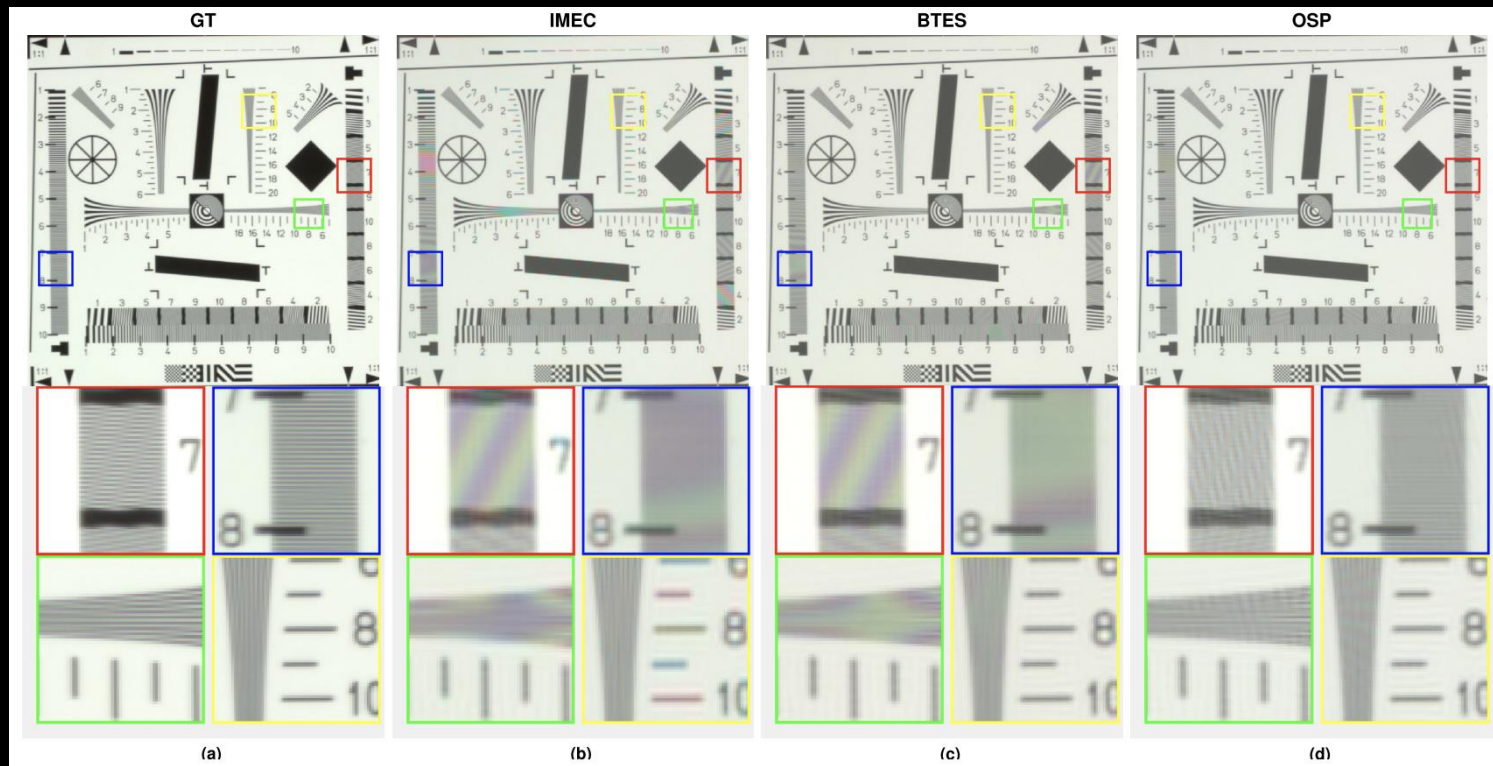
OSP

9	12	15	2	5	8	11	14	1	4	7	10	13	16	3	6
14	1	4	7	10	13	16	3	6	9	12	15	2	5	8	11
3	6	9	12	15	2	5	8	11	14	1	4	7	10	13	16
8	11	14	1	4	7	10	13	16	3	6	9	12	15	2	5
13	16	3	6	9	12	15	2	5	8	11	14	1	4	7	10
2	5	8	11	14	1	4	7	10	13	16	3	6	9	12	15
7	10	13	16	3	6	9	12	15	2	5	8	11	14	1	4
12	15	2	5	8	11	14	1	4	7	10	13	16	3	6	9
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16	3	6	9	12	15	2	5	8	11	14	1	4	7	10	13
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10	13	16	3	6	9	12	15	2	5	8	11	14	1	4	7
15	2	5	8	11	14	1	4	7	10	13	16	3	6	9	12
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Sphere Packing

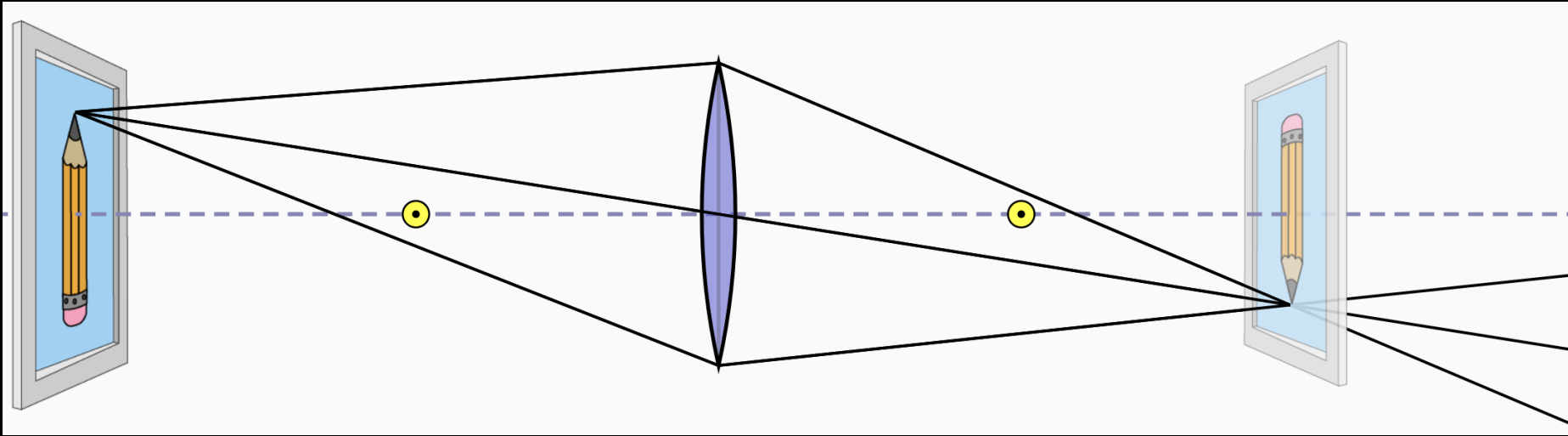


Reconstruction Results using MSFA



N. Diaz, A. Alvarado, P. Meza, F. Guzmán and E. Vera, "Multispectral Filter Array Design by Optimal Sphere Packing," in *IEEE Transactions on Image Processing*, vol. 32, pp. 3634-3649, 2023

Imaging



Pinhole Camera

